How the System Works

A plain-language guide to what was built, and what it is built from.

The salon runs on a website customers use to book and pay for appointments, and a private area where the two artists manage their day. Everything is online — there is nothing to install, and no computer in the salon that has to be switched on for it to work.

THE THREE PARTS

Almost every website of this kind is built from the same three pieces. It helps to think of the salon itself:

	WHAT IT DOES	BUILT WITH
The shop front	The pages people actually see and click: the home page, the service menu, the booking calendar, the artists' dashboards. It runs inside the visitor's own web browser.	React
The back office	The part nobody sees. It decides what is allowed — whether a time is free, whether a payment cleared, who may see which bookings — and it writes everything down. The shop front asks it questions; it answers.	Node.js & Express
The filing cabinet	Where the information itself lives: customers, services, appointments, payments, and each artist's opening hours. Hosted by MongoDB, who look after the backups and the hardware.	MongoDB Atlas

Keeping these apart is what makes the system safe. The rules live in the back office, not in the pages — so a customer cannot book a time that is already taken by editing what is on their own screen. The browser can only ask; the back office decides.

TWO OUTSIDE SERVICES

Two jobs are handed to specialists rather than built from scratch, because both are things you want a regulated company to be responsible for:

	WHAT IT DOES	ROLE
PayFast	Takes the R100 booking fee. Card details are typed on PayFast's own page and never pass through the salon's system, which is deliberate — the salon never holds anybody's card number.	Card payments
Gmail	Sends every email the system produces: sign-in codes, booking confirmations, cancellations, and the reminder an hour before an appointment.	Email

What it is built from

The same list again, with the names you would see on an invoice or a CV. Every one of these is free, widely used, and well documented — none of it is unusual or hard to hand over to somebody else later.

TECHNOLOGY	WHAT IT IS FOR
React	The library the visible pages are built with. It is what lets a page update the moment something changes — picking a service refreshes the list of free times without reloading the whole page.
Node.js	Lets the back office be written in the same language as the pages (JavaScript), so one developer can work on both sides.
Express	Organises the back office into clear routes — one address for bookings, one for payments, one for reports — so each request goes to the right place.
MongoDB & Mongoose	The database, and the layer that checks information is sensible before it is saved. An appointment without a service or a date is rejected rather than quietly stored.
Recharts	Draws the graphs on the Reports page.
PDFKit	Turns a report into a PDF the artists can download or print.
Nodemailer	Sends the emails, with the salon's own branded header at the top of each one.
Vercel	Hosts both halves and puts them online. Publishing an update takes about a minute and does not take the site down.
Jest	Runs 160 automated checks over the booking rules — see the last page.

SIGNING IN WITHOUT A PASSWORD

There are no passwords anywhere in the system. Signing in emails a six-digit code which is typed into the page, and that code works once, for five minutes, with three attempts before it is thrown away and a fresh one has to be requested.

This was a deliberate change. An earlier version offered three numbers to choose from, which meant a stranger with somebody's email address could simply try all three. Typing a code out of a six-figure range removes the guess entirely — and there is no password to be forgotten, reused, or stolen from elsewhere.

What happens when somebody books

The clearest way to see how the parts work together is to follow one booking from start to finish.

1. **They sign in.** They enter their email, receive a code, and type it in. If they have never booked before they create a profile first — name, email, phone — and confirm the email address is theirs.

2. **They choose a service.** Fourteen services across hair, nails, makeup and lashes, each with its own length, from a 30-minute lash application to a two-hour set.

3. **The system works out which times are genuinely free.** This is the part that does the most thinking, and it is explained on its own below.

4. **They confirm and pay.** The booking is held while they pay the R100 fee on PayFast's page. Nothing is confirmed until PayFast tells the system the money cleared.

5. **Everyone is told.** The customer gets a confirmation email with the service, date and time; the artist sees the appointment on their calendar; and an hour before it, the customer gets a reminder.

> **STEP 3, IN DETAIL — THE PART THAT IS EASY TO GET WRONG**
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> The booking page does not simply list the salon's opening hours. For each possible start time it checks four things: that the artist who does *that* service is free then; that she is free for the *whole* length of it, without a break in the middle; that it finishes by closing time at 17:00; and that the appointment is at least a day away, because the salon does not take same-day bookings.
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> A time only appears if all four hold. That is why a two-hour service never shows 16:00 — there is only one hour left in the day — and why today never appears at all. The same four checks run again in the back office when the booking is submitted, in case two people were looking at the same time slot at once.

TWO ARTISTS, TWO DIARIES

The salon has a nail artist and a hair, makeup and lashes artist, and the system knows which services each of them does. Their availability is kept separate, so one artist's open morning is never offered as availability for the other's services — and each sees only her own appointments on her dashboard, not the whole salon's.

What the artists can do

SECTION	WHAT IT IS FOR
Calendar	Set the hours they are available, week by week, and see what has been booked into them. Times outside business hours or in the past cannot be entered.
Bookings	Every appointment for the services they offer, newest first, with its status.
Dashboard	Today's or this week's schedule at a glance, plus recent history.
Reports	Four reports — busiest times, most popular services, bookings and cancellations, and new against returning customers — over any date range they choose, each downloadable as a PDF.
Masterclasses	Applications from people wanting to attend a masterclass.
Profile	Their own details, and which services they offer.

How it is kept correct

RULES THE SYSTEM WILL NOT LET YOU BREAK

Some rules matter enough to be enforced rather than trusted: no same-day bookings; nothing outside 08:00–17:00; the salon closed on Sundays; one customer cannot book two services that overlap in time; a booking fee cannot be paid twice; and an artist cannot book as a customer from their own account. Each of these is checked by the back office, so it holds however the request arrives.

AUTOMATED CHECKS

160 automated checks run over the booking rules before any update is published — the kind of thing that is tedious to test by hand and easy to break by accident. They cover whether a service fits in a gap, whether an email address is valid, what a report's date range means, and who is allowed where.

PUBLISHING AN UPDATE

Every change is recorded with a note explaining why it was made, so the history of the system can be read back. Publishing sends the current version to Vercel, which puts it live in about a minute; the previous version is kept, so a bad change can be reversed.

In short: an ordinary, well-understood set of tools, arranged so that the rules of the salon live in one place — the back office — where they cannot be sidestepped from a browser.